

Multi-Objective Reinforcement Learning for Sustainable Investment Strategies using FinRL

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Abstract. Modern investment decisions increasingly consider not only financial return, but also objectives such as risk minimization and sustainability. This work proposes a multi-objective reinforcement learning (MORL) framework built on FinRL to model sustainable investment strategies. The trading environment is adapted to output a reward vector composed of (i) financial return, (ii) risk penalty, and (iii) ESG sustainability score. Scalarization is applied to transform this vector into a single reward signal, and agents are trained using the PPO algorithm. To explore diverse trade-offs among the objectives, we incorporate the NSGA-III algorithm to evolve a population of scalarization weights and generate a Pareto front of optimal policies. The results show that MORL agents can outperform traditional PPO and mean-variance optimization (MVO) strategies in terms of long-term performance, risk control, and ethical alignment. This highlights the potential of combining deep reinforcement learning and evolutionary optimization to create flexible and responsible investment systems.

1 Introduction

Modern investment strategies are no longer driven solely by the pursuit of high returns. Instead, they increasingly incorporate additional objectives such as risk management and sustainability. These scenarios are naturally modeled using *Multi-Objective Markov Decision Processes* (MOMDPs). However, many practical reinforcement learning (RL) frameworks, including FinRL, are focused on scalar reward functions.

In this work, we adapt FinRL to support multi-objective reinforcement learning. We introduce a reward vector combining:

- Financial gain (r_1),
- Risk or volatility penalty (r_2),
- ESG-based sustainability score (r_3).

This adaptation enables experimentation with weighted preferences over multiple investment goals, allowing better alignment with real-world investor priorities.

In this work, we contribute a practical integration of Multi-Objective Reinforcement Learning (MORL) into FinRL. We adapt the original environment to output vectorized rewards, capturing financial return (r_1), risk penalty (r_2), and sustainability score (r_3). These are scalarized using weights λ_1 , λ_2 , and λ_3 to reflect different investor preferences.

We train policies with PPO under varying scalarization preferences and evolve these weight combinations using NSGA-III. This

evolutionary process identifies a Pareto front of trade-offs, from which representative policies are selected and evaluated on real financial data. This allows us not only to generate a set of diverse strategies, but also to compare them directly against classical PPO, mean-variance optimization, and the Dow Jones Index.

Our results show that policies guided by multi-objective optimization can achieve superior or more balanced performance across objectives compared to baseline methods. This highlights the potential of MORL for realistic financial decision-making where multiple criteria must be considered simultaneously.

2 Objectives

General objective

Design and evaluate a multi-objective reinforcement learning system in FinRL capable of generating investment policies that balance financial return, risk, and sustainability.

Specific objectives

- Adapt the FinRL framework to support vector-valued rewards combining return (r_1), risk penalty (r_2), and ESG sustainability score (r_3).
- Integrate scalarization techniques to enable training with customizable investor preferences through weights λ_1 , λ_2 , and λ_3 .
- Implement and apply the NSGA-III algorithm to discover a diverse set of scalarization weights representing Pareto-optimal trade-offs.
- Train and evaluate policies under selected weight configurations derived from NSGA-III and compare their performance against classical PPO, MVO, and benchmark indices.
- Analyze how different preference configurations influence portfolio behavior and objective satisfaction.

3 Methodology

1. Analyze and understand FinRL's trading environment and baseline PPO agent implementation.
2. Select a diversified portfolio of assets and enrich the dataset with ESG scores, using real indicators or simulated proxies where needed.
3. Redesign the reward function to emit a 3-dimensional vector composed of:
 - r_1 : financial return,
 - r_2 : volatility-based risk penalty,

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- r_3 : ESG-based sustainability score.
4. Apply linear scalarization to combine the objectives into a scalar reward: $r = \lambda_1 r_1 + \lambda_2 r_2 + \lambda_3 r_3$.
 5. Train agents with PPO under both manually defined and automatically optimized scalarization weights.
 6. Integrate the NSGA-III algorithm to evolve a population of $(\lambda_1, \lambda_2, \lambda_3)$ vectors and approximate the Pareto front of optimal policies.
 7. Evaluate the trained policies in a realistic out-of-sample test environment, recording their performance on each objective.
 8. Compare the performance of NSGA-III-derived strategies with classic PPO, mean-variance optimization (MVO), and the Dow Jones Industrial Average (DJI) benchmark.
 9. Visualize the learned Pareto front and analyze trade-offs exhibited by selected strategies under different investor preferences.

4 Algorithms, Data and Simulators

• Algorithms:

- **Proximal Policy Optimization (PPO)**: chosen as the primary reinforcement learning algorithm due to its robustness and stability in continuous action spaces.
- **NSGA-III (Non-dominated Sorting Genetic Algorithm III)**: used to optimize scalarization weights $(\lambda_1, \lambda_2, \lambda_3)$ and discover a diverse set of Pareto-efficient policies balancing return, risk, and sustainability.

• Data:

- Historical stock prices (2009–2025) for companies such as Apple, Microsoft, Amazon, and others, sourced via Yahoo Finance.
- ESG scores approximated via a simulated proxy dictionary, inspired by publicly available sustainability reports and research methodologies.

• ESG Integration and Reward Design:

This project adopts the conceptual framework of the **Sustainalytics ESG Risk Score**, which evaluates a company’s exposure to material environmental, social, and governance risks:

- The ESG score is defined on a continuous scale where **lower values indicate lower ESG risk** (i.e., better sustainability performance).
- As FinRL does not provide real ESG data for all assets, we simulate ESG scores normalized between $[0, 1]$ for each asset.
- These scores are integrated into the environment via a configurable dictionary.

At each timestep t , the environment outputs a reward vector:

$$\vec{r}_t = \begin{bmatrix} r_{\text{return}} \\ r_{\text{risk}} \\ r_{\text{ESG}} \end{bmatrix} \quad \text{which is scalarized as:}$$

$$r_t = \lambda_1 r_{\text{return}} + \lambda_2 r_{\text{risk}} + \lambda_3 r_{\text{ESG}}$$

This allows for flexible agent training under different investor profiles, and facilitates the use of NSGA-III to discover policies that lie on the Pareto front of objective trade-offs.

• Simulator:

- A modified version of FinRL’s `StockTradingEnv`, adapted to:

- * Return 3-dimensional vector rewards,
- * Accept $(\lambda_1, \lambda_2, \lambda_3)$ as dynamic configuration parameters,
- * Log individual reward components for evaluation and visualization.

5 Results

To evaluate the effectiveness of our multi-objective reinforcement learning framework, we compare the performance of various investment strategies using the evolution of portfolio value across a fixed testing period.

Figure 1 shows the portfolio trajectories of four approaches: classical PPO (in blue), Mean-Variance Optimization (MVO, in orange), the Dow Jones Industrial Average (DJI, in green), and our proposed Multi-PPO agent trained with scalarized multi-objective rewards (in magenta). The Multi-PPO strategy consistently outperforms all others, reaching higher cumulative returns while maintaining relative stability. Notably, the MVO strategy achieves strong gains but exhibits higher volatility, while PPO and DJI remain more conservative.

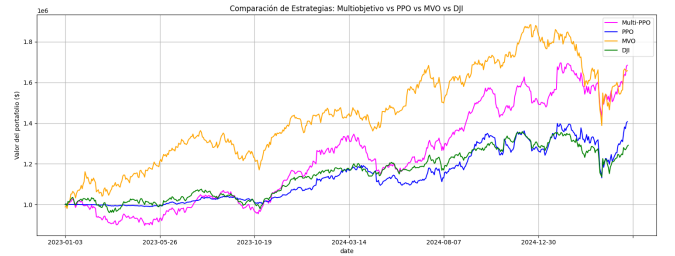


Figure 1. Comparison of PPO, Multi-PPO, MVO, and DJI.

To explore the impact of different scalarization weights $(\lambda_1, \lambda_2, \lambda_3)$, we employed NSGA-III to discover a Pareto front of non-dominated policies, each offering a distinct balance between financial return, risk, and sustainability. Figure 2 presents the results of three selected policies obtained from this Pareto set. Each curve reflects the performance of a specific configuration:

- The purple policy $(1.95, 0.61, 2.03)$ prioritizes both return and ESG and delivers the best overall performance.
- The brown policy $(1.95, 0.96, 2.06)$ offers a well-balanced approach with moderate risk mitigation.
- The red policy $(1.03, 0.13, 1.93)$ focuses more on ESG and stability, showing lower volatility but comparatively slower growth.

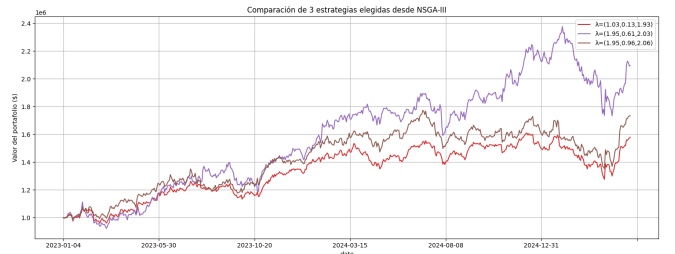


Figure 2. Comparison of three policies selected from the NSGA-III Pareto front with different λ values.

These trajectories highlight the adaptability of our framework: by adjusting the scalarization vector λ , investors can tailor policies to emphasize profitability, risk reduction, or ESG performance according to their preferences.

Finally, Figure 3 visualizes the complete Pareto front discovered by NSGA-III in a three-objective space. Each point corresponds to the average performance (r_1, r_2, r_3) of a trained policy. The color scale reflects the return value (r_1), making it easier to identify high-performing solutions. Multiple perspectives help visualize the shape of the Pareto surface and expose trade-offs. Some solutions offer higher returns at the expense of increased risk or lower ESG compliance, while others prioritize sustainability and risk control with moderate returns.

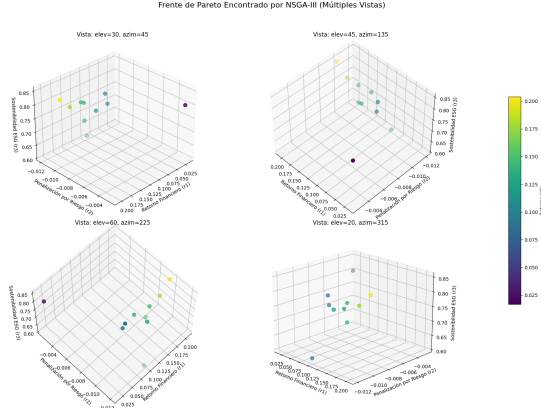


Figure 3. Pareto front discovered by NSGA-III, visualized from multiple angles. Each point represents a policy and its average performance on return (r_1), risk penalty (r_2), and ESG score (r_3).

Together, these figures confirm the utility of multi-objective reinforcement learning and Pareto-based search in constructing adaptable and effective investment strategies. The scalarized reward design and NSGA-III optimization enable the discovery of diverse solutions, empowering investors to align portfolio behavior with complex and evolving priorities.

6 Discussion

The results obtained in this project highlight the promising role of multi-objective reinforcement learning (MORL) in financial portfolio management. By modifying the FinRL environment to output vector-valued rewards encompassing financial return, risk, and ESG sustainability, we enabled the learning agent to internalize more complex, realistic investment objectives.

Initial experiments used hand-picked scalarization weights ($\lambda_1, \lambda_2, \lambda_3$) to combine the reward components into a single training signal. While these configurations already delivered competitive results—particularly outperforming classical PPO and closely rivaling MVO strategies—the approach lacked a systematic mechanism to explore different investor priorities.

To address this, we introduced NSGA-III as a meta-optimizer over the scalarization weights. Instead of selecting a single λ configuration, the algorithm evolved a diverse set of candidate policies optimized across the three objectives. The resulting Pareto front, visualized from multiple perspectives in Figure 3, confirmed that the objectives were in trade-off: improving financial return often came at the cost of either increased risk or lower sustainability.

By selecting three representative policies from this Pareto set (Figure 2), we demonstrated how adjusting the λ values yields distinctly different portfolio trajectories. For example, one configuration led to a highly profitable yet volatile strategy, while another favored consistent growth with reduced ESG exposure. This illustrates the practical flexibility of our MORL framework to adapt to varying investor risk profiles and ethical considerations.

Moreover, when compared to traditional baselines (Figure 1), the Multi-PPO agent consistently delivered superior or comparable returns while incorporating additional ESG and risk-awareness dimensions. This supports the hypothesis that reward vectorization combined with evolutionary optimization can produce robust and diversified strategies.

In summary, our approach offers a data-driven toolkit to personalize portfolio construction along multiple axes of interest. Future work may focus on integrating real ESG scores, expanding the reward function to include other risk metrics (e.g., drawdown or Value-at-Risk), and extending the scalarization model beyond linear combinations to capture nonlinear investor preferences.

7 Conclusion

This project showcased the integration of Multi-Objective Reinforcement Learning (MORL) into the FinRL framework to train agents that optimize not only financial return, but also risk and ESG sustainability.

By scalarizing vectorial rewards, we explored both manually defined and automatically optimized weight configurations. NSGA-III enabled the discovery of Pareto-optimal strategies that balance investor preferences.

The resulting agents demonstrated diverse behaviors and competitive performance, often surpassing traditional baselines such as PPO, MVO, and the DJI index. This highlights MORL’s value in building ethical, personalized, and robust trading systems.

8 Dependencies and Setup

To ensure the proper execution of the proposed multi-objective reinforcement learning system, the following dependencies and configurations are required:

Python Version

- Python 3.10 (recommended)

Python Libraries

The following libraries must be installed via `pip` or `conda`:

- `numpy`
- `pandas`
- `matplotlib`
- `scipy`
- `stable-baselines3` → for PPO agent training
- `gymnasium` or `gym` (FinRL uses OpenAI Gym environments)
- `yfinance` → to fetch stock data from Yahoo Finance
- `finrl` → financial RL environment and baseline code
- `torch` (PyTorch) → deep learning backend

Optional Tools

- `tensorboard` → for monitoring training logs
- `seaborn` → for enhanced plotting (optional)

Environment Setup

You may create a virtual environment using Anaconda or `venv`:

```
conda create -n proyecto_RL python=3.10
conda activate proyecto_RL
pip install numpy pandas matplotlib scipy
        torch gym stable-baselines3 yfinance finrl
```

Custom Code Requirements

- The environment `StockTradingMultiEnv` is a modified version of `FinRL's StockTradingEnv`, adapted to return vectorial rewards and accept scalarization weights.
- Ensure all relevant scripts (training, NSGA-III, evaluation) are in the same working directory or properly imported as modules.
- A file named `resultados_nsga3.csv` is expected for policy evaluation with selected scalarization weights.

Hardware Requirements

While CPU execution is sufficient for experimentation, training with NSGA-III over many generations or policies may benefit significantly from GPU acceleration. PyTorch will automatically use GPU if available.

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